

Assignment 1: CSE 421

Your Name: Mehedi Hasan Rowdro

Section: 25

March 13, 2026

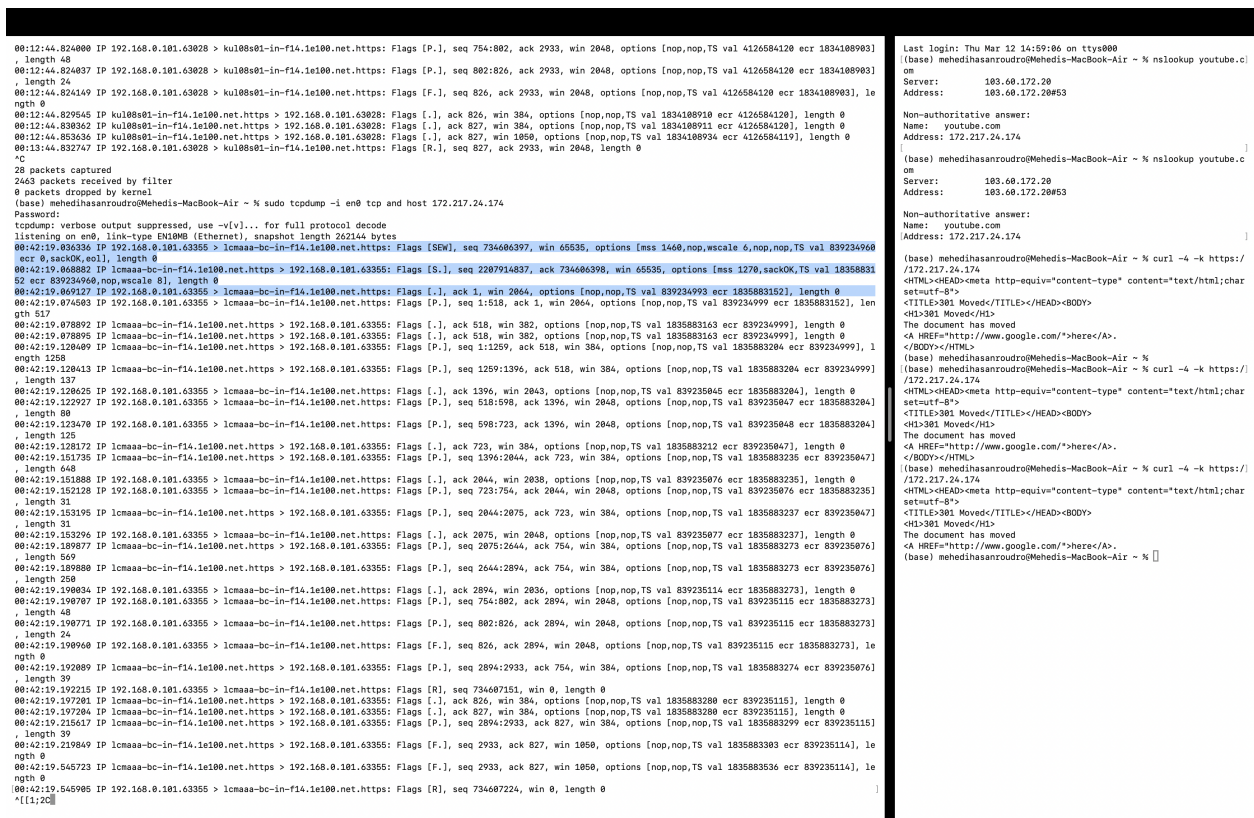


Figure 1: 3 Way

1 TCP 3-Way Handshake

**** *add the screenshot of the shell showing the connection establishment entries*

SYN Packet

Src Port: 63355	Dst Port: 443
Seq No: 734606397	
Ack No: 0	
Flags: SYN	Win: 65535
Options: mss 1460,nop,wscale 6,nop,nop,TS val 839234960 ecr 0,sackOK,eol	

SYN-ACK Packet

Src Port: 443	Dst Port: 63355
Seq No: 2207914837	
Ack No: 734606398	
Flags: SYN, ACK	Win: 65535

Options: mss 1270,sackOK,TS val 1835883152 ecr 839234960,nop,wscale 8

ACK Packet

Src Port: 63355	Dst Port: 443
Seq No: 734606398	
Ack No: 2207914838	
Flags: ACK	Win: 2064
Options: nop,nop,TS val 839234999 ecr 1835883152	

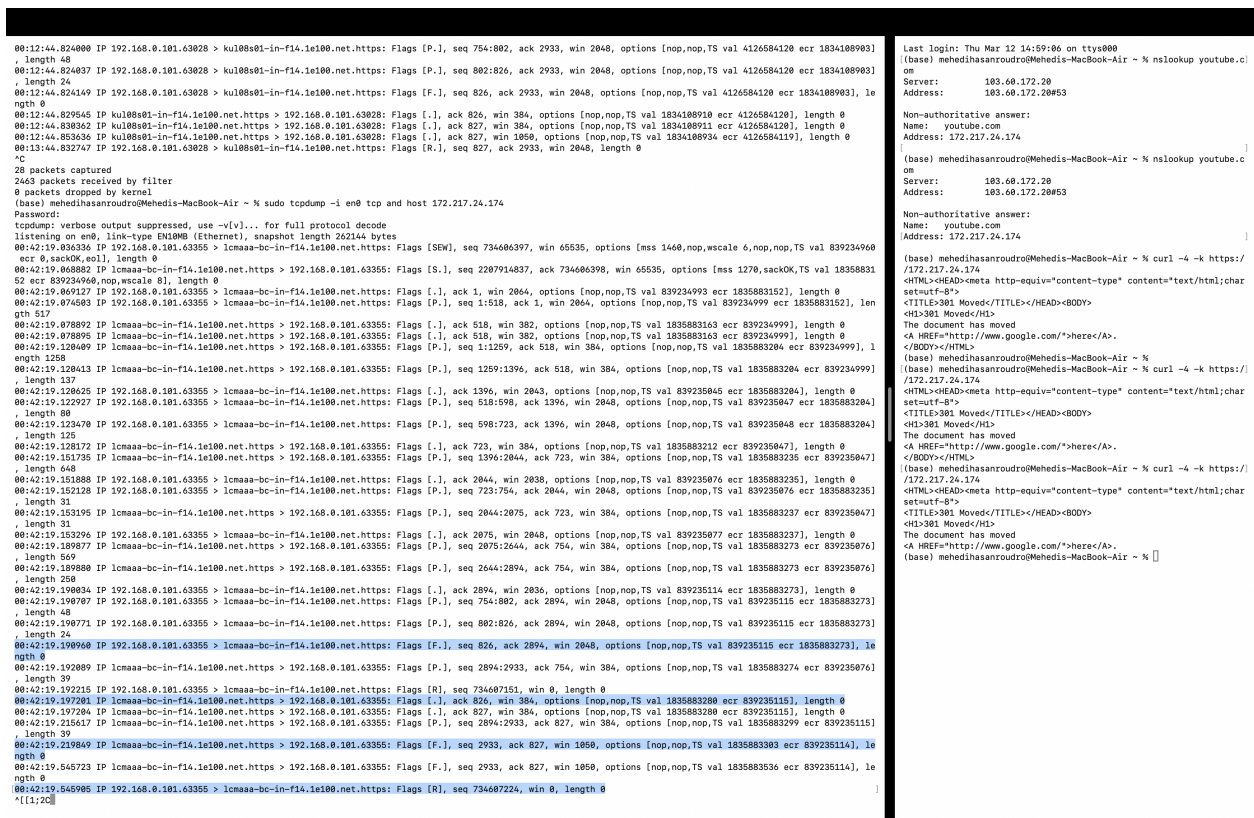


Figure 2: 4 Way

2 TCP 4-Way Connection Termination

**** *add the screenshot of the shell showing the connection termination entries*

FIN Packet (from client)

Src Port: 63355	Dst Port: 443
	Seq No: 826
	Ack No: 2894
Flags: FIN, ACK	Win: 2048
Options: nop,nop,TS val 839235115 ecr 1835883273	

ACK Packet (from server)

Src Port: 443	Dst Port: 63355
	Seq No:
	Ack No: 826
Flags: ACK	Win: 384

Options: nop,nop,TS val 1835883280 ecr 839235115
--

FIN Packet (from server)

Src Port: 443	Dst Port: 63355
Seq No: 2933	
Ack No: 827	
Flags: FIN, ACK	Win: 1050
Options: nop,nop,TS val 1835883303 ecr 839235114	

ACK Packet (from client)

Src Port: 63355	Dst Port: 443
Seq No: 734607224	
Ack No:	
Flags: RST	Win: 0
Options: none	

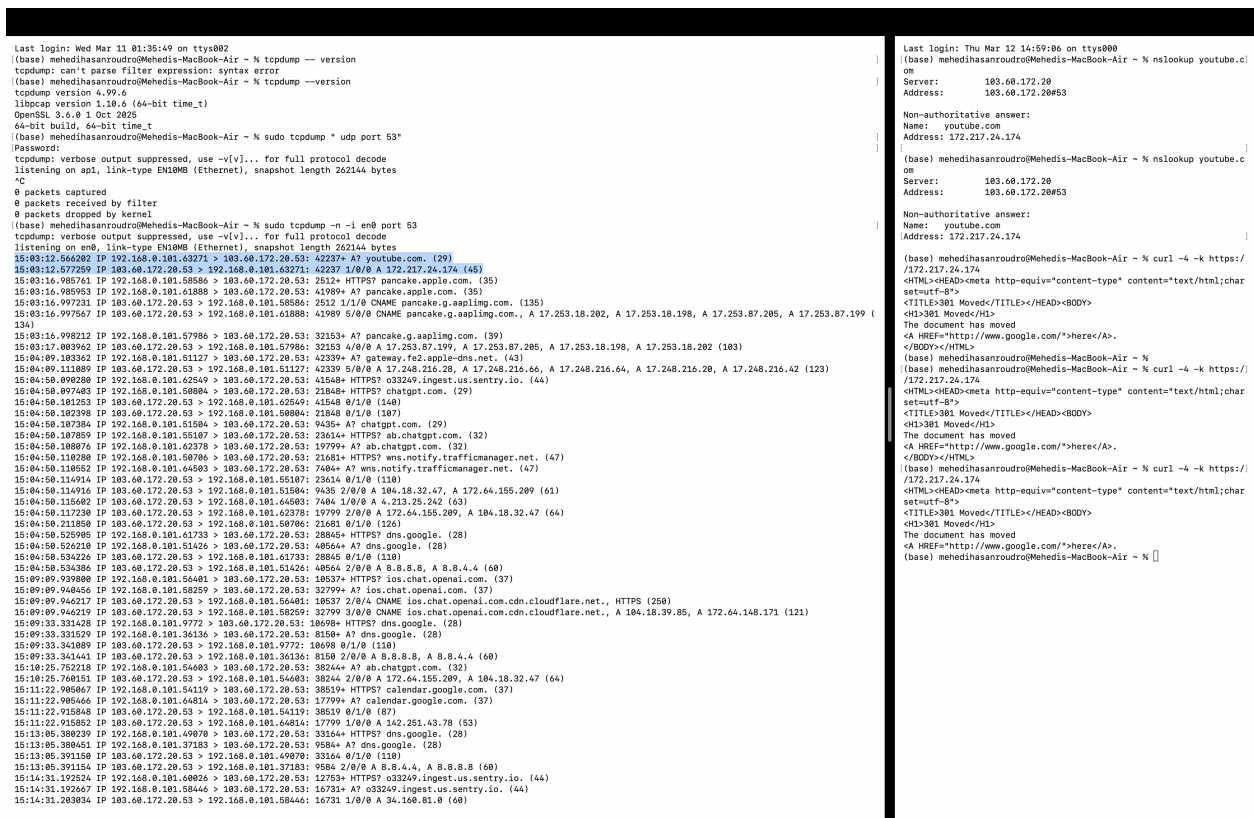


Figure 3: DNS

3 DNS Resolution

**** *add the screenshot of the shell showing the DNS request and response entries*

- Domain selected: youtube.com

DNS Query Packet

Field	Value / Explanation
Source Port	63271
Destination Port	53
Transaction ID	42237
Query Type	A
Query Name	youtube.com
Flags	recursion desired

DNS Response Packet

Field	Value / Explanation
Source Port	53
Destination Port	63271
Transaction ID	42237
Response Type	Answer
Response Data	172.217.24.174
Flags	(e.g., response, recursion available, authoritative) response, recursion available